

Memory, Forgetting and Measurement Error

Surv/SurvMeth 632

Aaron Maitland

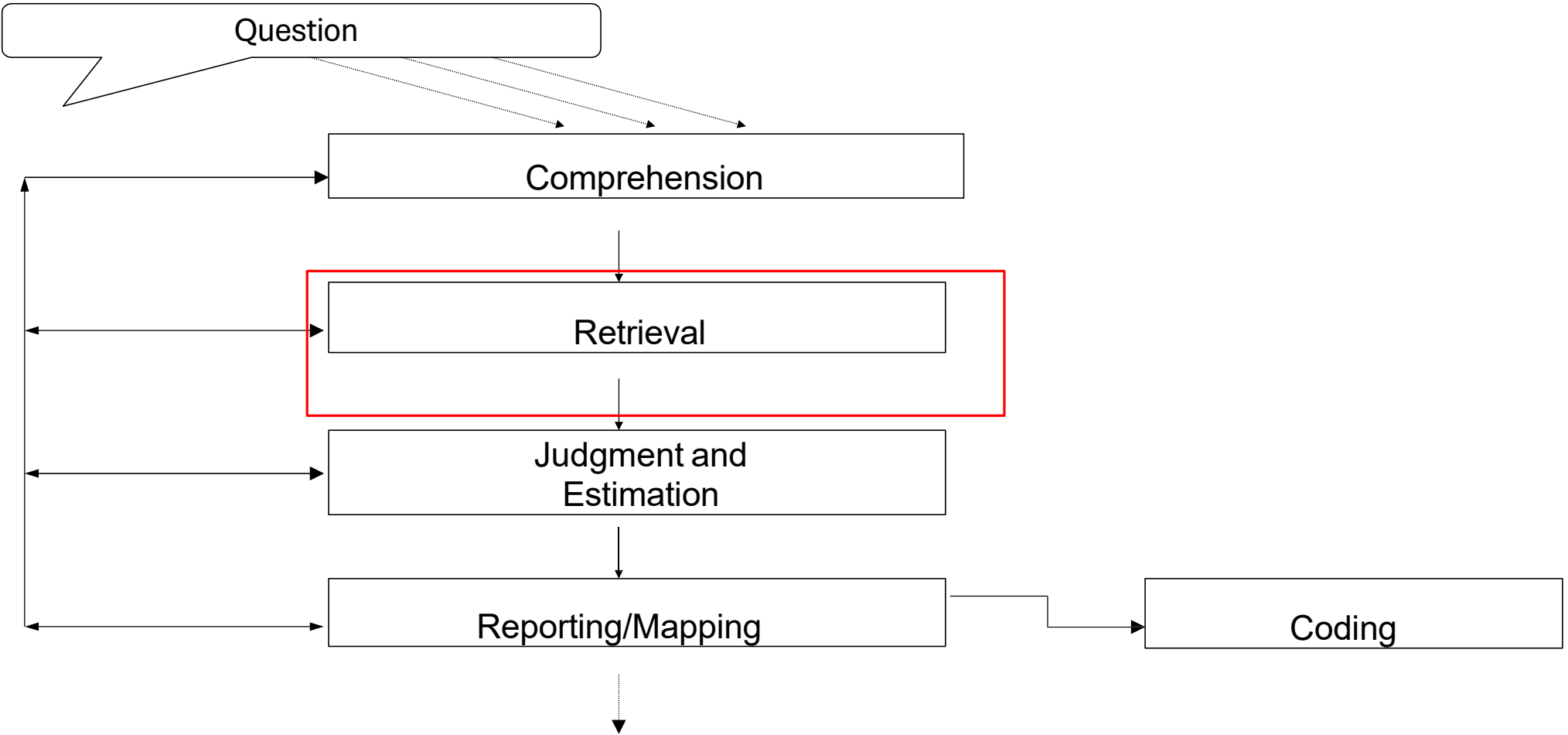
September 18, 2024

Outline: Sources of memory error

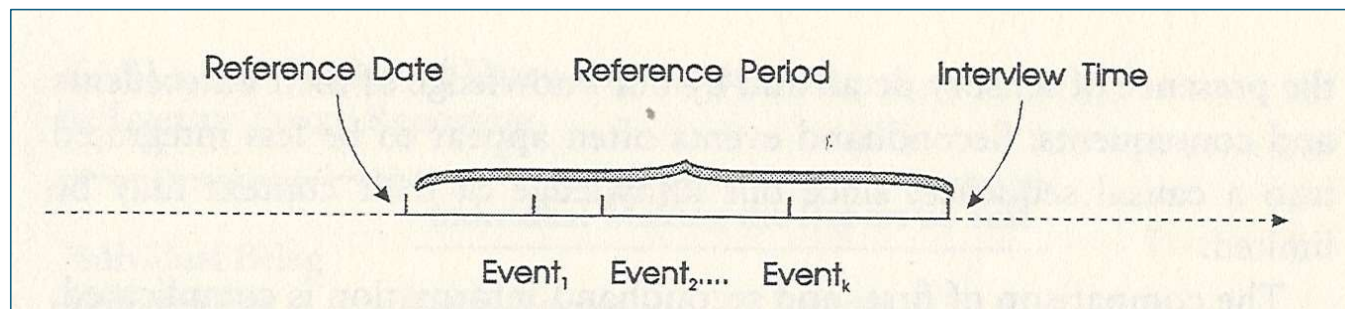
A: Basic Processes

1. Types of Memory
2. Encoding problems
3. Similarity and distinctiveness
4. Changing what is remembered
5. Delay

B: Memory Improvement



Some key concepts for memory retrieval and survey questions



- During the past 12 months, that is, since September of 2023, ABOUT how many days did illness or injury keep you in bed more than half the day? [NHIS]
- As of the end of LAST WEEK, how long had you been looking for work? [CPS]

Memory is an imperfect record

- Forgetting
 - Fundamental to everyday notion of human memory
 - contrasts with computer memory where no forgetting
 - Typically assumed to result from decay and/or interference
 - Clearly, a source of survey measurement error, whatever the mechanism
- Processes that affect recall accuracy
 - Incomplete encoding
 - Similarity of events
 - Changes to memories over time:
 - elaboration, remembered inferences
 - Delay, i.e., passage of time

Different Kinds of Memory

- Long Term Memory
 - Episodic – memory for autobiographical events
 - Semantic – knowledge of the world, including oneself
 - Procedural – involves performance of actions or routines like riding a bike
- Working memory is “mental scratch pad”
 - limited capacity storage, e.g., for spoken words as they arrive, for partial products in an estimation task
 - Preserves order information (or allows it to be reconstructed)
- Retrieval is the process of bringing information from LTM to an active state to be used
 - When information reaches sufficient level of activation it enters working memory (Baddeley, 1986)
 - Activation spreads from one concept or memory in LTM to related ones
- “...memory for experience is intertwined with general knowledge; similarly, the process of remembering what actually happened is inextricably bound up with the process of inferring what probably happened.” (Tourangeau, Rips, and Rasinski, 2000, p. 97)

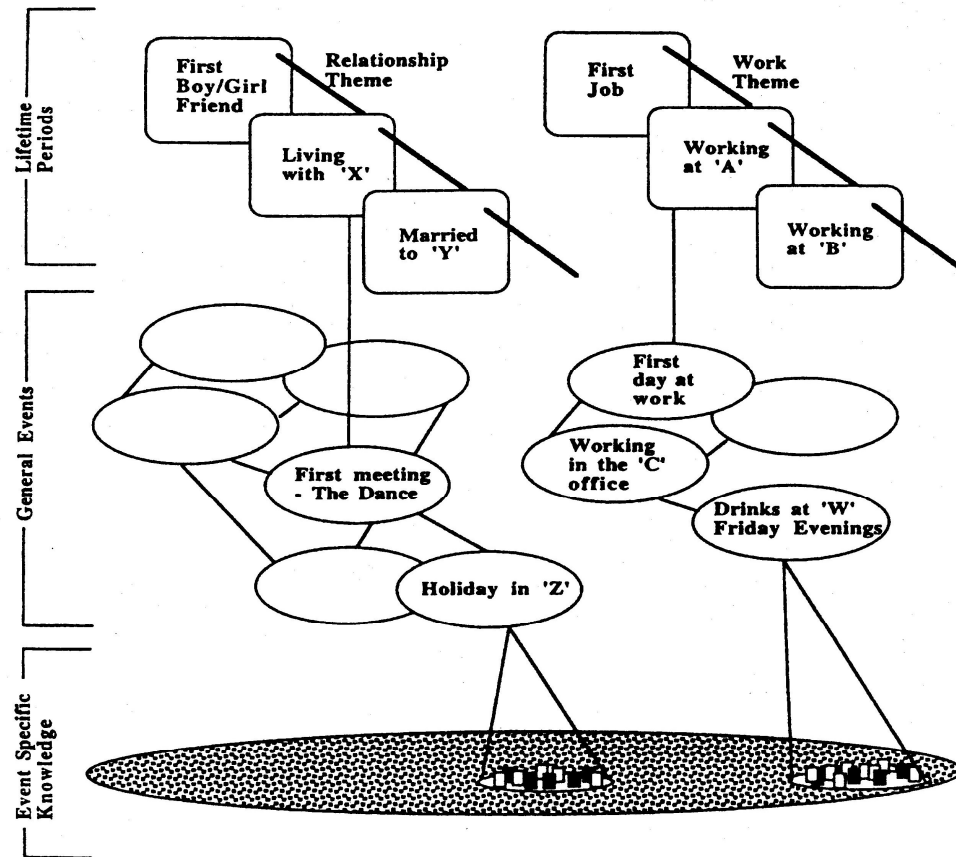
Episodic versus semantic memory

- Episodic (autobiographical) memory
 - memory for events – the everyday notion of memory
 - autobiographical events are basic units
 - forgetting is a normal part of the process
- In everyday life, episodic memory is “autobiographical”
 - Memory for the events and periods of one’s life (e.g., Thomsen, 2015)
- Supplemented by semantic memory
 - knowledge about the world
 - e.g., trees have leaves, chiropractors are not MDs
 - includes general knowledge about oneself (though not specific events)
 - e.g., my birth date, I shop for groceries once a week, I had my appendix removed when I was a young kid
 - can be used to infer what is absent in our (episodic) memory for events
 - Forgetting content of semantic memory is not normal -- could indicate pathology

Three models of memory (Tourangeau, Rips, and Rasinski, 2000)

- Tulving
 - Episodic memory: knowledge of autobiographical events
 - Semantic memory: generic knowledge
- Kolodner's model
 - Single system that represents both generic properties of similar events and the properties that distinguish one event from similar events
 - Event memory organization packets (E-MOPs) represent types of events and include general information about the events like participants, location, larger events, and more general event classes
 - Retrieval process generates indices to distinguish events
- Conway's multilayered model
 - Lifetime periods: discrete periods with identifiable temporal boundaries
 - General events: extended autobiographical sequences and generic events
 - Event specific knowledge: details about general events

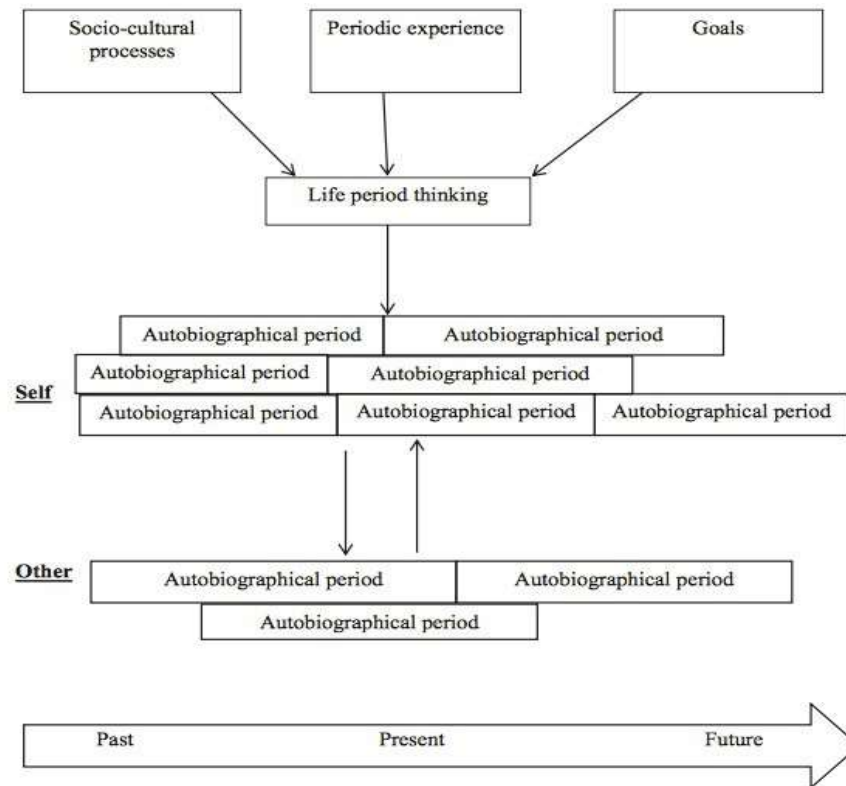
Conway's model of autobiographical memory



Autobiographical Periods

- Thomsen (2015) gives autobiographical periods (APs) central role in autobiographical memory:
 - APs are temporal categories that contain information about emotion, typical people, places, objects, and activities associated with the period as well as temporal and causal relations between associated memories
 - More abstract than specific memories
 - Can last from a few days (e.g., a summer holiday) to decades (e.g., a marriage)
 - empirically, mean is 4 to 15 years
 - APs can facilitate retrieval and dating of specific events that occur in those periods
 - APs are serial, parallel, hierarchical; people can anticipate future APs
 - Specific events from the start and end of an AP are better recalled than events from elsewhere in the period

Autobiographical Periods



Thomsen, 2015 (figure 2)

A few take-aways from Conway's and Thomsen's models

- Recall of events and their dates is better when search involves Autobiographical/Lifetime periods, as opposed to calendar years
- Events from beginning and end of period remembered better than events from middle
- The combination/intersection of periods can help with reconstructing dates
 - – e.g., “it was while I was in college, while dating Brian, but before my job as a bartender”

Failure to encode detail

- For events to be remembered they must be “encoded,” i.e., recorded in people’s memories
- Lee et al. (1999) demonstrated that parents remember little about children’s vaccinations because do not encode these events
 - compared parents’ reports to medical records
 - reported “up-to-date” for 80% of children who are not up-to-date
 - recall is not improved by calendars or show cards (experiment 1)
 - recall is poor immediately after injection and no worse after 10 weeks (experiment 2)
 - recognition is no better than recall (experiment 3)
 - childhood injections are frequent, not particularly distinctive, occur in batches and involve esoteric terms
 - Parents may simply have not encoded enough information to accurately recall and report the events

Encoding and limits of recall: Depth of Processing

- When encoding occurs, more and less memorable attributes encoded
 - According to Craik and Lockhart (1972) memory retention is affected by the depth of cognitive processing.
 - Craik & Tulving (1975) asked participants about different aspects of words: visual* (V), auditory (A), semantic (S)
 - V: Is this word in upper or lower case? TABLE
 - A: Does this word rhyme with “dog”? Log
 - S: Does the word field fit into this sentence? The horse lived in a _____?
 - Then tested recognition of same words
 - More superficially encoded material (visual) less likely to be later recognized than more deeply encoded (semantic) material
 - Recall accuracy: $V < A < S$
 - Richer, more elaborated associations from semantic processing led to better recognition
 - The more deeply processed a stimulus (event) the better it will be recalled

*typeface characteristics

Encoding and the Limits of Recall: Encoding Specificity

- When context of encoding and retrieval match each other, memory better than when differ
- Tulving & Thomson, 1973:
 1. Participants studied word pairs e.g., train-black
 - responsible for 2nd word, i.e., the “study word”; train is “context” for black
 2. Participants presented word strongly associated with study word, e.g., white, and asked to produce 4 associates possibly if included study word (black) then context differed between study (train) and generation of associate (white)
 3. Participants asked if recognized any associates from study phase
 - when generated the study word, e.g., black, only recognized 54%

Encoding Specificity (2)

4. Then presented first word from pairs (“context word”), e.g., train, and asked to recall second word:
 - 61% of second words were recalled – better than recognition (which is almost never the case)
 - Of these, 42% had not been recognized in previous phase!
- Take-away: When context at encoding and retrieval match, recall substantially improved
- In fact, better than recognition when out of context

Encoding specificity (3)

- Encoding specificity demonstrated across a range of contexts
 - Physical context
 - Godden & Baddeley (1975) asked divers to study words on shore or 20 feet under
 - Tested their recall for the words in either same or different context
 - Divers showed vastly superior memory when recalled in same context as studied
 - “Context dependent learning”
 - Mental state
 - Eich, et al. (1975) instructed participants to smoke either marijuana or tobacco before studying words and smoke either one or the other before testing
 - Participants showed superior memory when smoked same substance before study and test

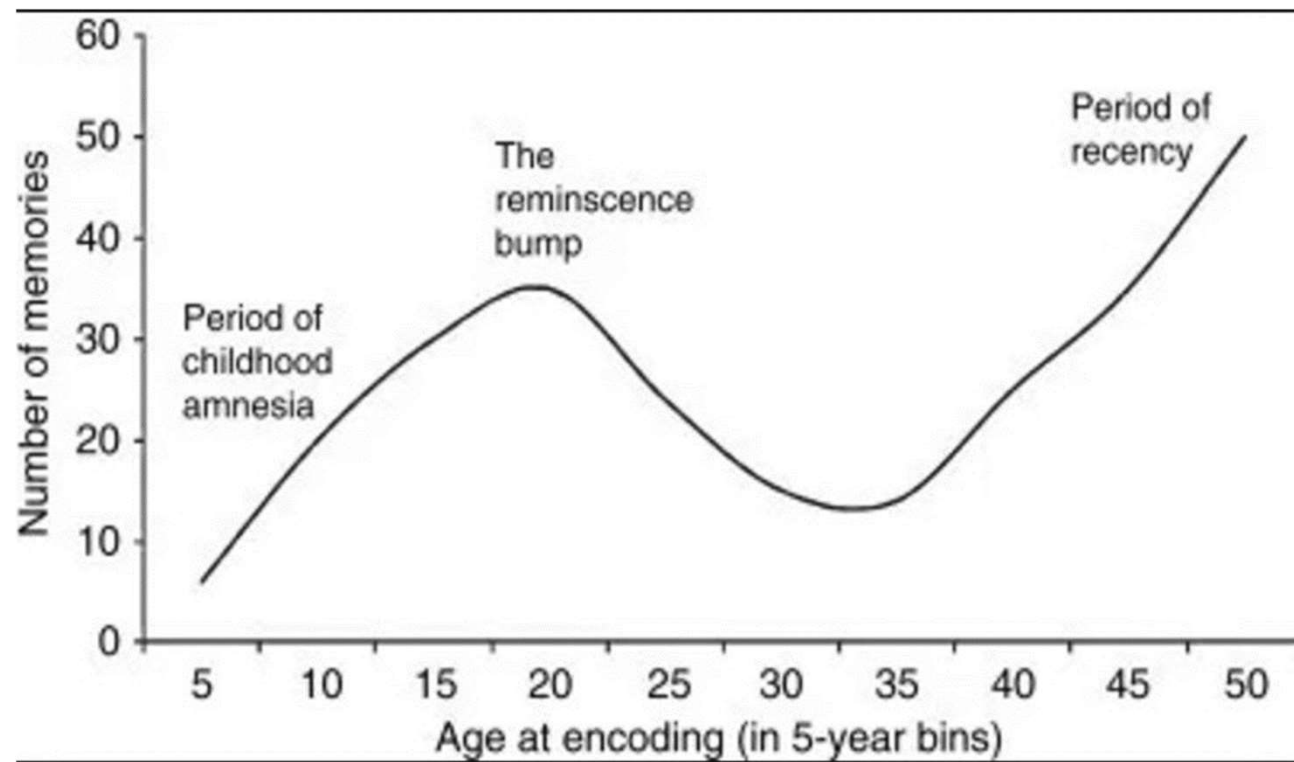
Encoding specificity (4)

- Emotional state
 - Bower, Monteiro & Gilligan (1978) hypnotically induced either positive or negative state when studying a list of words and induced either state before testing
 - Subjects' recall was best when study and test states matched
 - Failure to replicate (1985)
- Mood Disorder
 - Participants high in “trait anxiety” better recalled negative emotion words (Eysenck & Byrne, 1994) than participants low in anxiety
 - Attributed to more anxious participants focusing more on negative events and thus encoding them more thoroughly
 - Because anxiety is often a chronic state, encoding and recall contexts match for negative words for participants high in anxiety

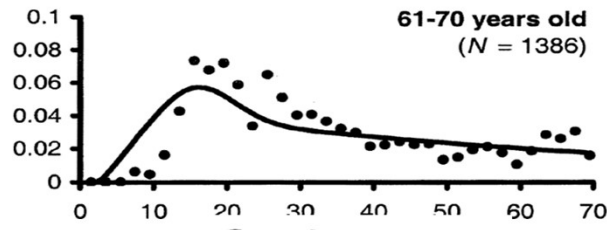
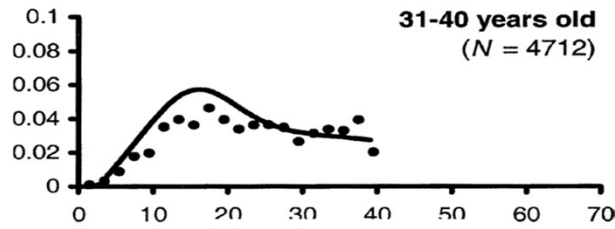
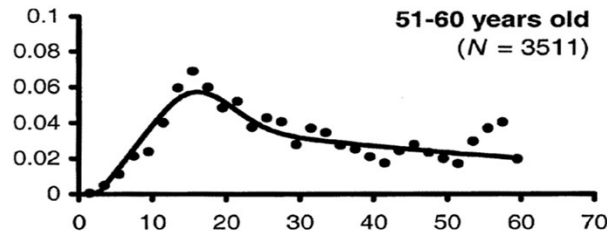
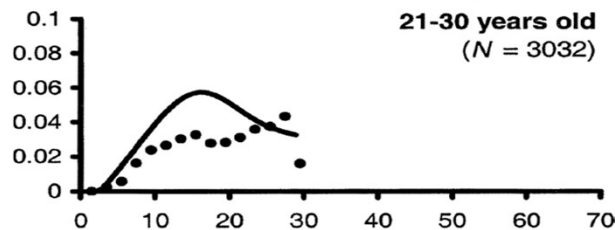
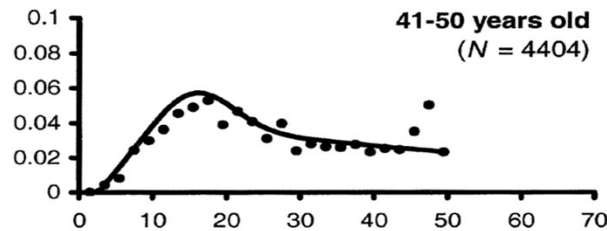
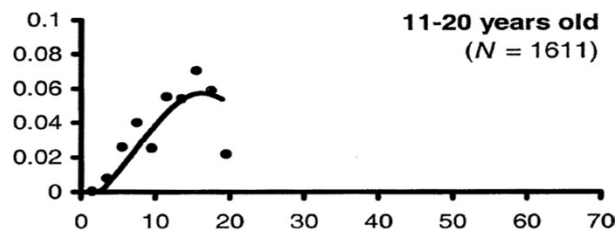
More Events Encoded in Some Life Periods than Others: The Reminiscence Bump

- When people recall over the lifespan, tendency to recall more events from ages 10-30
 - Attributed to more effective encoding during times of change
 - Events from this period have special status in autobiographical memory
- Poor recall prior to this age attributed to “childhood amnesia”
- Poor recall after this age attributed to ordinary forgetting
- Improved recall near present likely due to recency

Reminiscence Bump



Reminiscence Bump most clear after 40



Subject age

Subject age

Jannsen, Chessa & Murre (2005).

Memories elicited with Galton-Crovitz word cue method

Example

Cue word: car

"I remember the time I bought my first car. I had to save up to afford it. It was a red Toyota. I was 22 years old."

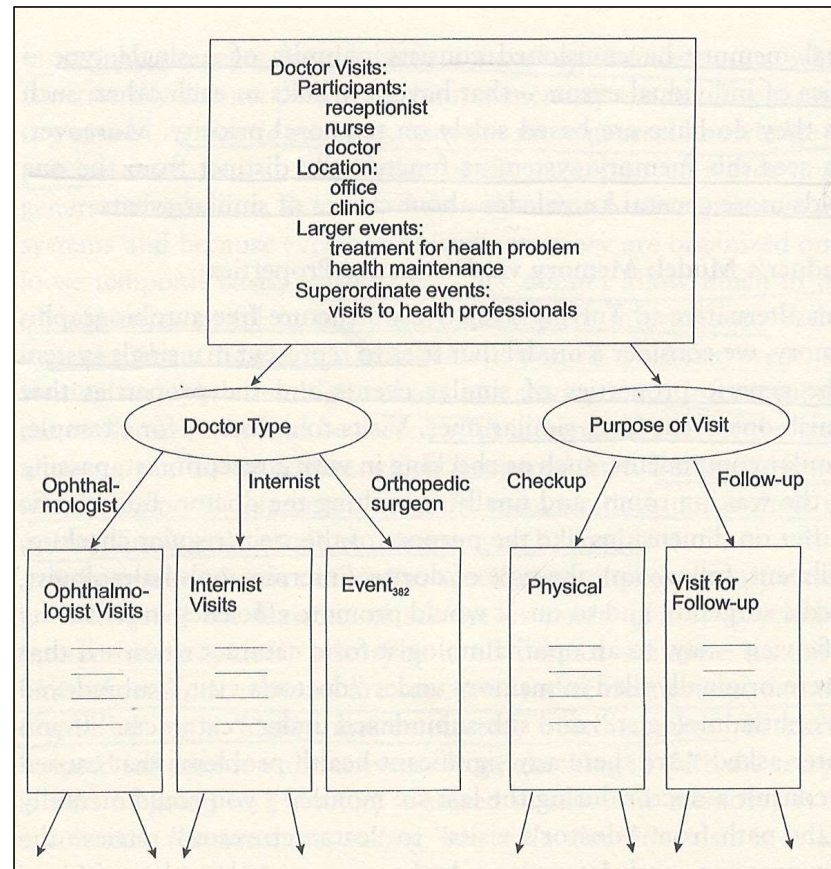
Similarity and distinctiveness

- More difficult to recall events when the to-be-recalled event is similar to other stored events
- Similar events can become blended into a single generic memory
 - Would need to “unpack” the generic memory to report component events
 - easier to remember common than distinguishing features
- Leads survey respondents to abandon conventional recall-and-count strategies
 - More about this later in the course when we discuss estimating frequency

Similarity and Distinctiveness (2)

- Schank and Abelson's script (or schema) idea consists, essentially, of slots for the common features of event types
 - e.g., doctor visits often involve talking to a receptionist, waiting in a waiting room, being examined by a clinical assistant or nurse, then being examined by the physician
- Distinguishing features of individual events (or episodes) must be remembered separately
 - e.g., doctor extremely young

Kolodner's model (CYRUS) and similarity of events



Tourangeau, Rips, and Rasinski, 2000, p.74

Similarity and Distinctiveness (3)

- Are generic memories harmful for data quality?
 - Yes:
 - Means & Loftus (1991) found worse recall for recurring than isolated medical visits (e.g., allergist for regular treatments vs. orthopedist for broken bone
 - Recall of recurring doctor visits more likely to involve generic memories
 - No:
 - Smith (1991) found advantage for generic memories in recall of dietary intake when reference period is longer:
 - There is a better chance of eating what one usually eats – generic memory – over more meals
 - Using diaries as gold standard, accuracy better (lower intrusion rate) for longer retention intervals

Similarity and Distinctiveness (4)

- Can individual events be recalled when generic memories exist?
 - Yes:
 - Means & Loftus (1991) improved recall of specific medical visits by asking about specialists and providing timeline
 - No:
 - Neither Linton (1982) nor Wagenaar (1986) could improve their own recall for individuating detail with repeated attempts
- Implications for writing questions that break up generic events?

Changing what is remembered

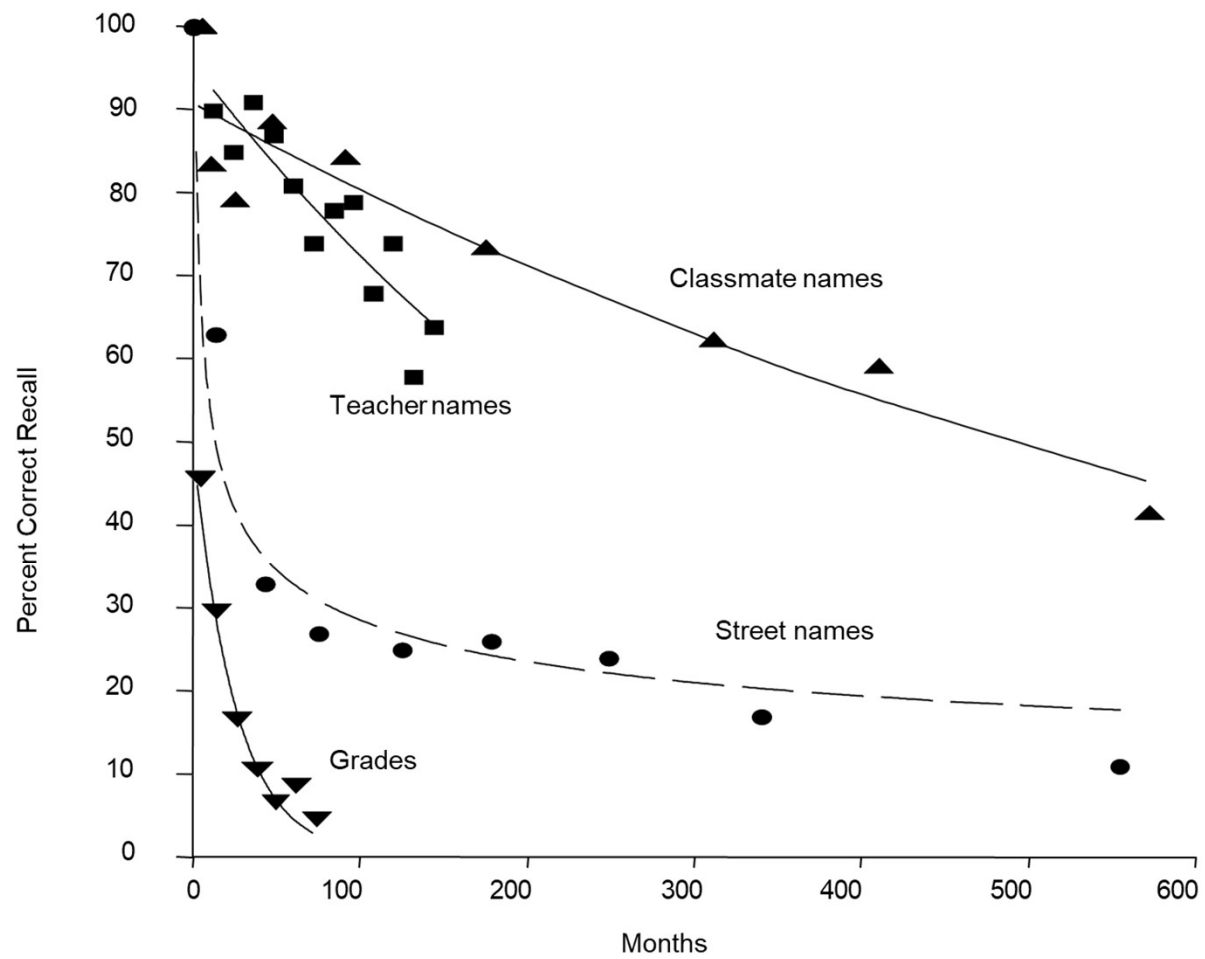
- Incorporating inferences into memory (Loftus, 1979)
 - Participants viewed film of traffic incident and were then asked:
 - “How fast was the car going when it went through the yield sign”
 - Led to reports of yield sign in original traffic event on a subsequent memory test even though a stop sign, not a yield sign, was present in the film
 - Explanation is that participants infer that the traffic sign (to which they did not pay attention) was a “yield” sign (because question asked about this); people do not seem to distinguish between inferred and experienced events
 - Source Amnesia

Changing what is remembered (2)

- Incorporating imagination into memory (Gary, Manning, Loftus & Sherman, 1996)
 - Adult participants rated likelihood they experienced 40 childhood events on scale from “definitely did not happen” to “definitely did happen”
 - Two weeks later, half of participants were asked to imagine some of these events for 1 minute – e.g., breaking window with hand, getting in trouble and calling 911
 - Later, asked to re-rate likelihood of having engaged in 40 events; authors analyzed % originally “unlikely” re-rated as “likely”
 - Change in this direction more common among participants who imagined events than among those who did not
 - For broken window event, 24% who imagined the event re-rated it as “likely” vs. 12% who did not imagine it
 - For 911, 20% versus 11%
- Also, source amnesia: imagined events are incorporated into memory for actual event; the fact that the event was imagined is forgotten

Effects of delay on recall

- Age of events dramatically affects their memorability
- Retention functions (theoretical forgetting curves) may vary with what is being recalled
- but most forgetting
 - is monotonic
 - continues as age of events increases
 - e.g., recall of classmate's names still degrading after 50 years!
 - Suggests something is still represented even when recall unsuccessful



Source: Tourangeau, Rips & Rasinski (2000)

Effects of delay on recall (cont'd)

- In surveys, age of recalled relevant events is usually function of length of recall (reference) period that question asks about
 - Neter & Waksberg (1964) observe that longer reference periods involve not only older events but more events
 - more events to forget and more interference from similar events
 - longer recall periods lead to decrease in number recalled events per unit time
 - Blair and Burton (1987) observed less recall-based responding when behavioral frequency questions had longer recall periods
 - Rs may recognize the limits of their memories, or
 - they may not be willing to exhaustively recall many events

Metamemory

- Knowledge about one's memory
 - Children get better at predicting memory performance as they get older
 - Older adults generally recognize their reduced memory ability but still overestimate performance
- Knowing that you don't know
 - Glucksberg & McCloskey (1981) showed that if no relevant information is stored, participants quickly respond "don't know"
 - Does Margaret Thatcher use an electric toothbrush?
 - But if potentially relevant facts are retrieved, they are examined in detail; if facts do not lead to an answer, slow "don't know"
 - Does Bert Parks have a degree in journalism?

Techniques that may improve recall

- More Time
- Decomposing the Task
- Richer Retrieval Cues
- Varying Recall Order
- (the other kind of) Cognitive Interviews

Caveats about memory improvement

- Results in literature are mixed
 - sometimes certain interventions appear to help, sometimes they do not
- Caveat: More is not necessarily better
 - interventions that increase number of reported events usually taken as evidence for improved recall
 - but may be increasing false reports/overreports

More time

- Facts retrieved with repeated recall attempts, long after initial recall
 - Williams and Hollan (1981) found people recall more high school classmates after nine hours (!) of recalling
- Implies Rs should be given more time to respond
 - Cannell et al. (1981) used longer Qs (without additional content) to slow interviews; increased # reports
- Longer response deadlines improve recall of some event categories, but not all (Burton & Blair, 1991)
 - may encourage recall when estimation more appropriate
- Implications for recall in different modes of data collection?

Decomposing the task

- Can help recall if break event into its components
 - Means et al. (1994) asked about cigarette smoking at certain times of day and in particular situations
 - more smoking reported with this technique than when respondents asked to give a “gut reaction” or use whatever strategy they chose
- But decomposition can lead to lower accuracy
 - can potentially lead to overreporting within subcategories
 - Belli, Schwarz, Singer & Talarico (2000) – will revisit later in the course

Retrieval Cues: Event History Calendars

- Event history calendars (EHCs)
 - Often a grid of periods by activities, completed in whatever order Rs find natural
 - Theory is that recall is facilitated by
 - Sequential cues: tap into Lifetime/Autobiographical periods
 - Parallel cues: connect contemporaneous events from different life domains, e.g., employment and relationships

Tobacco Life Events Calendar																						
Month		NOV2003			DEC2003			JAN2004			FEB2004			MAR2004			APR2004			Month		
	Third of Month	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3		Third of Month	
A	Landmarks																				Landmarks	
B	Residence																				Residence	
C	Living With																					
	Spouse																				Spouse	
	Partner																				Partner	
	Parents																				Parents	
	Housemates																				Housemates	
	Children																				Children	
D	Life Event																					
	Marital Status																				Marital Status	
	Pregnancies																				Pregnancies	
	Children Born																				Children Born	

Retrieval Cues: Event History Calendars (2)

- Belli, Shay & Stafford (2001)
 - Compared quality of reports from (1) EHC and (2) traditional questionnaire (Q-list) to Q-list reports by same respondents in prior year
 - Respondents in Panel Study of Income Dynamics (PSID)
 - Previous PSID responses serve as “gold” standard
 - Compared to Q-list, EHC produced better agreement with previous PSID responses for residential moves, income, unemployment, illness – so additional cues seem to help
 - produced overreporting in # of new household members, # of jobs
 - EHC did not take reliably longer than Q-list interviews

Data Quality Findings (Belli, et al. 2001)

	<u>Whether</u>	<u>Direct Quantity</u>	<u>Derived Quantity</u>	<u>When</u>
Move	EHC			
Entered Residence			ehc	
Left Residence			=	
Jobs		ehc		
Income		EHC		
Employed		=	ehc	EHC
Unemployed		EHC		
Out of Labor Force		=	ehc	EHC
Vacation		=		
Self-III		EHC		
Other-III		EHC		
ADC/AFDC	=		Q-L	=
Food Stamps	=		=	=

EHC = significantly better memory performance than Q-L; ehc = marginally better memory performance than Q-L; Q-L = significantly better memory performance than EHC

Retrieval Cues: Event History Calendars (3)

- Belli, Smith, Andreski & Agrawal (2007)
 - Compared EHC and conventional questionnaire (CQ) to PSID reports over previous 30 years
 - EHC led to more accurate reports for cohabitation, employment, unemployment and smoking history
 - CQ provided better data for marriage history
- Belli, Bilgen, Al Baghal (2013)
 - EHCs most helpful when retrieval task most difficult
 - e.g., Respondent whose work history is intermittent

Retrieval Cues: Types of Cues

- Some cues are better than others in cued recall tasks
 - Brewer (1988) and Wagenaar (1986) found that cues about what happened better than who or where, which are better than when
 - Barsalou (1988) found that locations prompt recall of more events (so are better cues) than activities which prompt more than broad time periods or participants
- In both sets of studies, date cues least helpful presumably because people do not “time stamp” events, i.e., dates not encoded with events

Varying recall order

- Mixed results
 - Loftus & Fathi (1985) found advantage for reverse order – start at present and work backward in time – for recall of exams in psychology class (similar for recalling teachers names, Whitten & Leonard. 1981)
 - Jobe et al. (1990) found small advantage for forward recall of dietary intake
 - Loftus, Smith, Klinger & Fiedler (1992) found no advantage for either strategy for health care visits
- If clear causal sequence/narrative structure, then early to late is probably better – but not established empirically yet

(The other kind of) cognitive interview

- Fisher & Quigley (1992)
 - Cognitive Interview (CI) technique from forensic psychology
 - Attempt to improve recall with context reinstatement, focused retrieval, extensive retrieval, varied retrieval, and multiple representations
 - Pilot study about dietary intake suggests advantage for CI over standard recall group
 - CI group recalled 12.2 items of food vs. 5.9 for standard group
 - with no increase in intrusions
 - and better recognition: higher hit rate and lower false alarm rate
 - Is this practical for surveys?

Implications for Designing Survey Questions

- Reference periods
 - Longer or shorter?
- Lag between recall period and data collection?
- Retrieval cues?
- Autobiographical periods and recall periods?

TABLE 3.2 Summary of Factors Affecting Recall

Variable	Finding	Implication for Survey Design
Characteristics of Event		
Time of occurrence	Events that happened long ago harder to recall	Shorten the reference period
Proximity to temporal boundaries	Events near significant temporal boundaries easier to recall	Use personal landmarks, life events calendars to promote recall
Distinctiveness	Distinctive events easier to recall	Tailor length of the reference period to properties of the target events; use multiple cues to single out individual events
Importance, emotional impact	Important, emotionally involving events easier to recall	Tailor length of the reference period to the properties of the target events
Question Characteristics		
Recall order	Backwards search may promote fuller recall	Not clear whether backward recall better in surveys
Number and type of cues	Multiple cues typically better than single cue; cues about the type of event (what) better than cues about participants or location (who or where), which are better than cues about time (when)	Provide multiple cues; use decomposition
Time on task	Taking more time improves recall	Use longer introductions to questions; slow pace of the interview

Tourangeau, Rips, and Rasinski, 2000, p.98